https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	diameter/*d* is the independent variable and frequency/*f* is the dependent variable **or** vary *d* and measure *f*	**1**
	keep *L* constant or length (of tube) constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • tube supported • (loud)speaker positioned in line with the tube • (loud)speaker labelled	**1**
	labelled microphone, positioned outside tube in line with tube, connected to labelled oscilloscope or correct circuit symbol	**1**
	adjust/change frequency until maximum amplitude detected	**1**
	use calipers to measure *d*	**1**

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Question	Answer	Marks
1	**Method of analysis**	
	plot a graph of $\frac{1}{f}$ against d **or** d against $\frac{1}{f}$ (Do not accept logarithmic graphs.)	**1**
	for $\frac{1}{f}$ against d $v = \frac{2L}{y\text{-intercept}}$ **or** for d against $\frac{1}{f}$ v = gradient $\times$ k **or** $v = -\frac{\text{gradient} \times 2L}{y\text{-intercept}}$	**1**
	for $\frac{1}{f}$ against d k = gradient $\times$ v **or** $k = \frac{\text{gradient} \times 2L}{y\text{-intercept}}$ **or** for d against $\frac{1}{f}$ $k = -\frac{2L}{y\text{-intercept}}$	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 wear ear defenders (to prevent damage to hearing/to avoid loud sounds) **or** use a low volume <u>to prevent damage to hearing</u>/<u>to avoid loud sounds</u>	
	D2 use a rule to measure *L*	
	D3 increase frequency from a <u>low</u> frequency to the <u>first</u> maximum amplitude	
	D4 method to determine *f* at maximum amplitude, e.g. increase frequency to *f*, then continue increasing frequency, and then decrease frequency until value of *f* determined	
	D5 method to determine period from oscilloscope, e.g. no. of divisions × time-base	
	D6 for frequency/time period determined by oscilloscope, $f = 1 / T$	
	D7 repeat measurements of *d* and average in <u>different</u> directions/positions or <u>along the tube</u>	
	D8 perform experiment in a quiet room	
	D9 signal generator connected to (loud)speaker in diagram	
	D10 relationship valid if a straight line produced (Do not accept through the origin.)	

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